Lag-Corrected Correlation Graphs for Pre-stimulus EEG Connectivity

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Abstract

Understanding inter-regional connectivity from short EEG windows requires choosing between connectivity measures whose tradeoffs aren't fully characterized. We develop a pipeline for constructing weighted electrode graphs from 500 ms windows in a temporally predictable and unpredictable paradigm. Our process integrates estimation of the temporal lag and correlation between electrodes, using numerical methods to compensate for nonlinearities encountered in our process. These observed parameters go on to inform force-directed drawing of graphs representing the estimated system structure. As validation, we show that force-directed graph drawings of our processed correlation and lag data approximate true scalp topology without being provided electrode coordinates, indicating that our pipeline captures genuine structure instead of noise.

1 Introduction

We sought to answer how temporal predictability (CISI vs VISI) modulates inter-regional connectivity between the auditory cue and tactile stimulus. Areas across the cortex coordinate during the gap, and the pattern of that coordination differs depending on whether the touch is reliably timed or not. EEG records the scalp-level signature of this coordination at millisecond resolution, which makes it a natural tool for asking how the network of cortical regions reshapes itself around a stimulus.

However, we quickly realized that answering this question requires choosing among connectivity measures with nuanced and non-obvious tradeoffs. A standard way to study that reshaping is to treat the EEG recording as a graph. Each electrode is a node, and there is some statistical relationship between electrode pairs representing an edge. The graph's topology can then be compared across conditions, time points, or trials. By taking a graph-theoretic approach such as this, we can observe the whole structure of the brain at a time and look for things like geometries or symmetries that are inherent in it. Hence, our investigation pivoted to attempting various methods at understanding and analyzing the EEG data across the time axis for multiple trials.

Research Question: How does the topological structure of neural connections adapt under a stimulus?

2 Methods

2.1 Dataset and Preprocessing

The EEG data came preprocessed with ICA artifact rejection. To prevent unnecessary variance, we examined only subject ID 5. We subsample only 30 trials from the P2 (VISI) condition for the 500 ms variants as we're analyzing the same anticipatory segment as the P1 (CISI) condition.

2.2 Analytical Framework

To make a simplified graph representation of the brain, we needed nodes and weighted edges. We could directly define the nodes as each electrode site, but there are many potential quantities that can be used as edge weights. Pearson correlation only captures similarity in amplitude fluctuations which limits the extent to which we can say a signal aligns with another. Thus, we considered a few options: Mutual Information (MI) could give us nonlinear dependence of two variables which can truly reveal information flow between electrodes, yet we have only 30 trials per condition which is insufficient to estimate MI. Phase Locking Value (PLV) was another option, but suffers from the same issues.

We ultimately decided to use correlation as our measure of coupling between electrodes. Although this statistic can't establish causality or prove directional communication between the neural regions, consistent similarity may indicate shared dynamics.

The correlation as given with our data cannot be calculated directly – we anticipated that there may be time lag or noise between the signal outputs of each electrode. To account for this, we chose to individually calculate the "optimal lag" between each channel pair based on the lag that minimizes the variance in correlation. Optimizing with respect to variance makes the correlation less likely to be a byproduct of noise and overall a stronger signal. Calculating the lag also allows us to establish a criterion for directionality, since a channel cannot be causing a signal in another node that is lagging ahead of it.

Then, between each channel we calculated the pairwise cross-correlation with the optimal lag applied between the signals averaged over the whole time period. Although this loses insight into the dynamics of the system during each trial, this allows us to observe how the general state of our electrode system changes as trials progress.

After cleaning our data, as discussed in the next subsection, we constructed a graph-based framework for representing electrode relationships. The graph contains 32 nodes, one for each electrode, with edge weights determined by thresholding correlation values. To visually analyze the structure of this graph, we needed to assign node positions that reflected the relative strengths of the connections. We used the ForceAtlas2[1] algorithm, which models the graph as a physical system: edges act like springs, with attraction proportional to edge weight, while nodes behave as repulsive bodies, similar to charged particles. The final graph layout is then given by a stable configuration of this system.

2.3 Statistical Assessment

The primary issue this approach introduces is that our correlation matrix with the applied lag is no longer positive semi-definite (PSD). The underlying covariance (non-normalized correlation) matrix, which generates our correlation matrix, is then also not PSD. This is because the lagging operation – effectively an infinite-dimensional operator – only left-multiplies our covariance matrix.

The typical solution of multiplying our covariance matrix on the right by this lag operator (which would be required to preserve the PSD nature of our matrix) would destroy the information that this operator represents, and is therefore not possible. Instead, we find the PSD covariance matrix that best captures the statistics of our lagged distribution's approximate (non-PSD) covariance matrix, and find the correlation matrix from this matrix.

The reason we examine the covariance matrix (rather than the correlation matrix we seek) is because our valid correlation matrices are members of the special linear group $\mathrm{SL}^+(32, \mathbb{R})$, while most of the useful transformations we could apply to our matrices do not respect the group structure. For example, transforming then forcibly re-normalizing a correlation matrix is inadvisable since the effect on the statistics would not be adequately represented by the entries of the correlation matrix. Additionally, we averaging over trials requires summation over correlation matrices which is in general not well-posed for matrices in our special linear group. Therefore we calculate correlation by first calculating (and subsequently processing) our covariance matrices, and only convert to a correlation matrix representation as the final step.

While our (real-valued) covariance matrices should in theory be real symmetric, we start by symmetrizing them in order to ensure floating point errors do not cause issues down the line. This is easily done by computing:

$$\mathbf{P}_{\text{symmetric}} = \mathbf{UT}^{\text{T}}(\mathbf{P}) + \mathbf{UT}(\mathbf{P}) - \text{Diag}(\mathbf{P}) \tag{1}$$

where $\mathbf{P}_{\text{symmetric}}$ is the symmetrized approximate covariance matrix, $\mathbf{UT}$ returns the upper triangular part of a matrix, Diag returns the diagonal part of a matrix, and $\mathbf{P}$ is the original covariance matrix.

Now that we have a symmetrized approximate matrix, we must re-project it to set of real positive semi-definite matrices, which form a convex cone in the vector space of all real symmetric matrices. This essentially represents the following nonlinear optimization problem:

$$\min_{\mathbf{P}_{\text{PSD}}} ||\mathbf{P}_{\text{PSD}} - \mathbf{P}_{\text{symmetric}}|| \tag{2}$$

Essentially, we project the $\mathbf{P}_{\text{symmetric}}$ matrix to the PSD cone by iteratively solving the matrix-valued optimization problem for the desired PSD matrix $\mathbf{P}_{\text{PSD}}$ which minimizes the distance $||\mathbf{P}_{\text{PSD}} - \mathbf{P}_{\text{symmetric}}||$ in the matrix-valued vector space. We chose to use the p-2 (Frobenius) norm to define our distance function. However, we found through experimentation that we could find a PSD matrix that approximates the optimal PSD matrix (to within less than a tenth of a percent at worst) through a single reprojection step without any further iteration needed.

Finally, we determined the average correlation by simply summing over N covariance matrices:

$$\mathbf{P}_{\text{sum}} = \sum_{i=1}^{N} \mathbf{P}_{\text{PSD}} \tag{3}$$

and then normalizing:

$$\mathbf{C}_{\text{PSD}} = \left[\sqrt{\mathbf{P}_{\text{PSD}}}\right]^{-1} \mathbf{P}_{\text{PSD}} \left[\sqrt{\mathbf{P}_{\text{PSD}}}\right]^{-1} \tag{4}$$

We can now utilize Marchenko-Pastur statistics to search for the significant eigenvalues in this $\mathbf{C}_{PSD}$ matrix, which is positive semi-definite and physically plausible. In order to avoid the same normalization and re-normalization problems we have already encountered, we do not directly remove the non-significant eigenvalues (those below the threshold implied by assuming Marchenko-Pastur statistics) from the correlation matrix. Instead, we remove the corresponding non-significant eigenvalues from the covariance matrix, leaving behind a covariance matrix that is still PSD and contains all of the information of the significant eigenvalues. We then normalize this filtered covariance matrix to generate the filtered correlation matrix:

$$\mathbf{SDS}^{-1} \equiv \mathbf{P}_{\text{PSD}} \tag{5}$$

where $\mathbf{D}$ is a diagonal matrix containing the eigenvalues λ_i of $\mathbf{P}_{\text{PSD}}$. Now define:

$$\delta(\lambda) = \begin{cases} 0 & \lambda \leq \varepsilon \\ \lambda & \lambda > \varepsilon \end{cases} \tag{6}$$

We use this to find a filtered diagonal matrix:

$$[\mathbf{D}_{diagonal}]_i^i = \delta(\lambda_i) \tag{7}$$

And from there the filtered correlation:

$$\mathbf{P}_{\text{filtered}} = \mathbf{S}\mathbf{D}_{\text{filtered}}\mathbf{S}^{-1} \tag{8}$$

$$\mathbf{C}_{\text{filtered}} = \left[\sqrt{\mathbf{P}_{\text{filtered}}}\right]^{-1} \mathbf{P}_{\text{filtered}} \left[\sqrt{\mathbf{P}_{\text{filtered}}}\right]^{-1} \tag{9}$$

Additionally, we calculated the matrix-valued difference between correlation matrices of subsequent trials (for both the CISI and VISI treatments) and calculated the p-2 norm of this difference. We then plotted this over trials to determine the best choice of number of trials to average over.

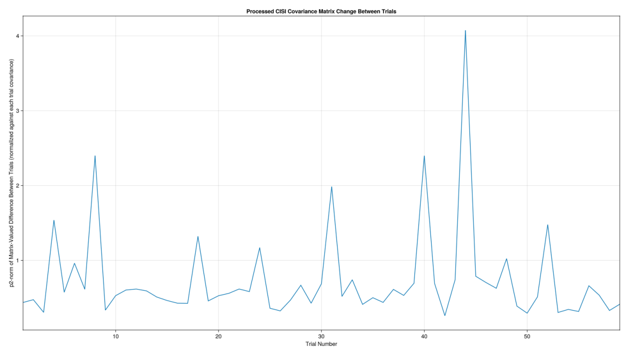


Figure 1: Magnitude of the change in the correlation matrix between subsequent trials of the P1 Condition (CISI)

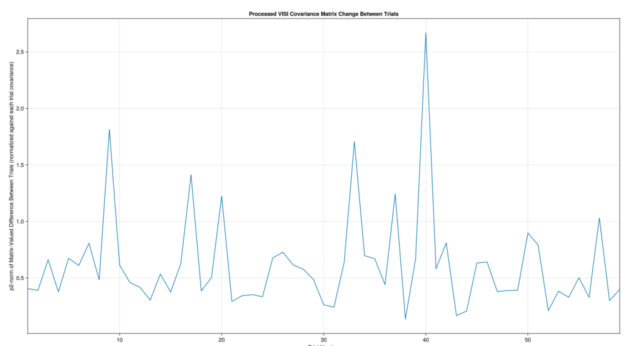


Figure 2: Magnitude of the change in the correlation matrix between subsequent trials of the P2 Condition (VISI)

Given these plots, an average over $N = 10$ trials was chosen.

3 Results

We ran anchor-based experiments and produced a few graphs comparing the 500 ms anticipation window between the cue and stimulus. We chose the C3 electrode as the anchor because the experiment had the tactile stimulus delivered to the right hand index, meaning the left central region of the sensorimotor area is a natural place to look at. Below are our figures for the two conditions:

From P1 to P2, we see a noticeable change in the frontal region fluctuation with relatively the same mean absolute value. This shows us that C3's relationship with frontal electrodes is more heterogeneous in the variable stimulus case which makes sense as the connections won't be as consistent as a predictable

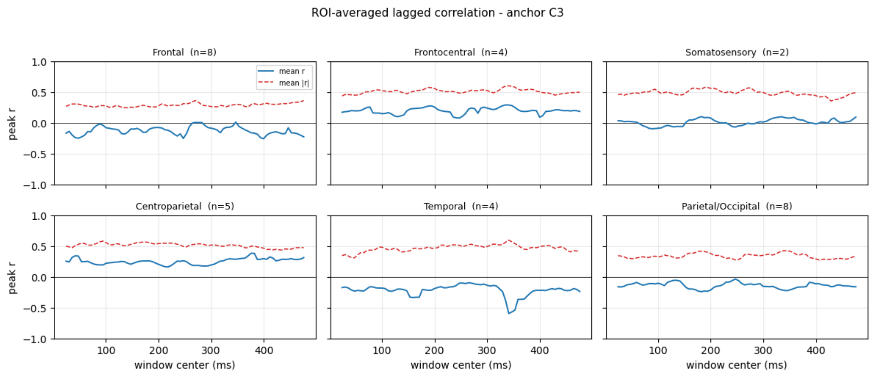


Figure 3: P1 Condition (CISI)

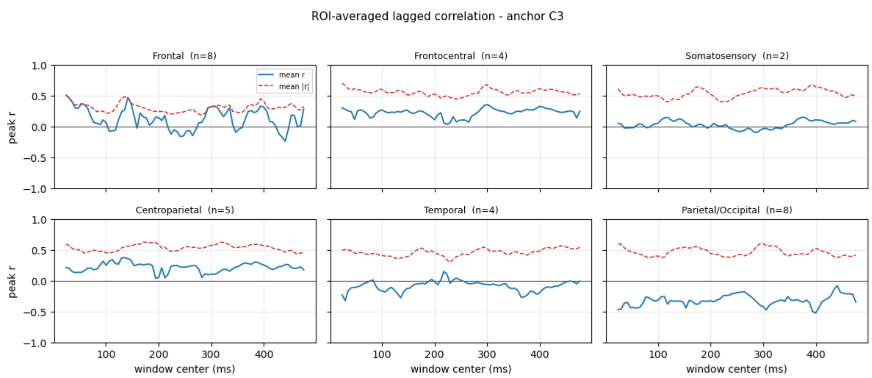


Figure 4: P2 Condition (VISI 500 ms variant subsampled)

second stimulus. The clearest positive coupling across both conditions appeared in frontocentral and centroparietal regions, while parietal/occipital channels showed a consistent negative relationship with C3. In contrast, nearby somatosensory electrodes showed weak signed average correlation despite moderate absolute correlation, suggesting canceling pairwise relationships within the ROI.

However, the above results should be taken with the following perspective: each graph has the mean |r| at 0.4-0.6 even when the mean r plot is close to zero most of the time. This shows that pairwise coupling is strong but their signs aren't consistent across the entire ROI, necessitating further electrode-electrode analysis.

We pursued a different visualization approach to get a picture of the lag's impact on the r-value across time. Here, we have three different graphs comparing the anchor (C3) with three different electrodes. For each graph, on the x-axis we have the window center in ms for each time point. On the y axis, we have the lag in ms where positive (above x=0) means anchor leads. The heatmap shows higher Pearson r values as red and lower as blue.

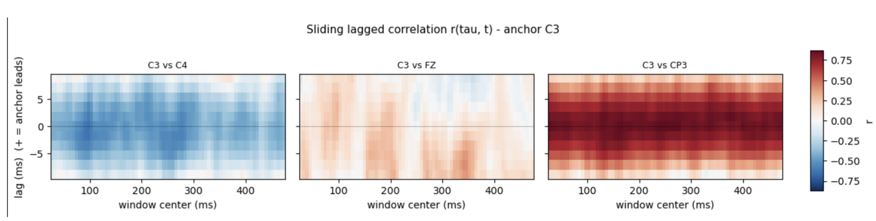


Figure 5: P1 Condition (CISI).

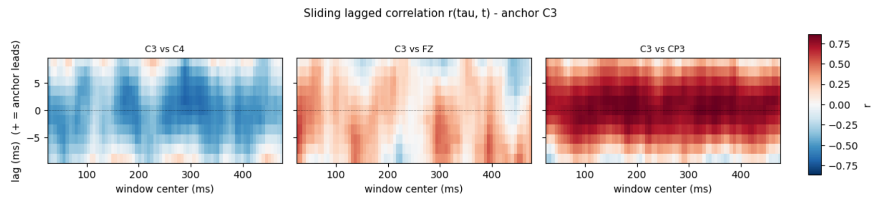
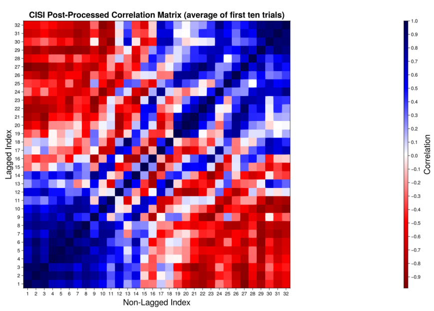


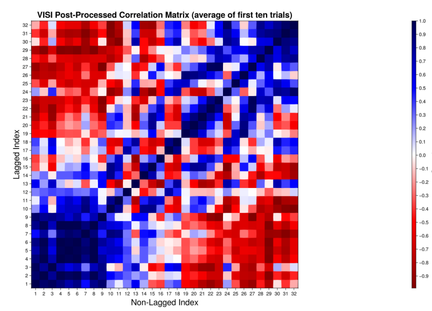
Figure 6: P2 Condition (VISI 500 ms variant subsampled.

The figures are quite similar across conditions with C3 and C4 correlating negatively across the entire timeline as they are are opposite sides of the midline; the C3 and FZ electrodes having a slightly positive correlation, and C3 and CP3 strongly correlating positively as they're nearby electrodes picking similar signals.

Now we will discuss the processed correlations obtained through our statistical analysis and using the lags mentioned above. It is worth noting that the starting condition varied between the CISI and VISI treatments; that is, the correlations averaged over the first ten trials of the CISI and VISI treatments respectively differed noticeably while still having a fairly similar overall structure.

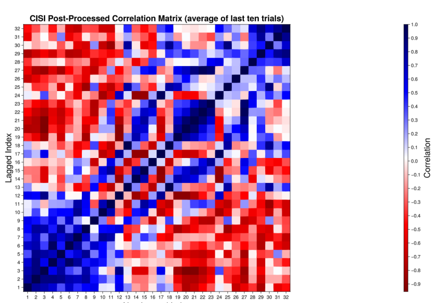


(a) Correlation in the first 10 trials of P1 (CISI) condition

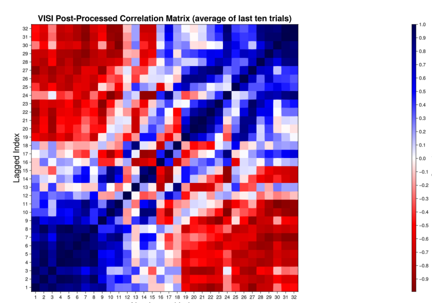


(b) Correlation in the first 10 trials of P2 (VISI) condition

At the end of both treatments, the correlations were once again averaged, this time over the last ten trials. Large-scale changes occurred in both cases, although much more in the CISI treatment. For instance, the scale of notable feature sizes in the VISI treatment correlation matrix has increased compared to the starting condition, while the structure of the CISI treatment correlation matrix has moved towards smaller feature scales compared to the starting condition. The physical significance of these changes will be discussed later.



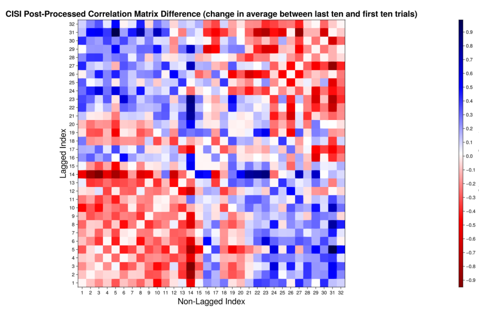
(a) Correlation in the last 10 trials of P1 (CISI) condition



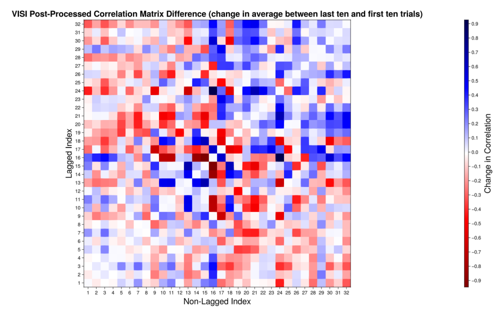
(b) Correlation in the last 10 trials of P2 (VISI) condition

The change in feature scale and structure can be seen more easily when the change between the starting and ending condition correlation is taken as a matrix difference. In the CISI treatment, distinct regions of the correlation matrix (corresponding to a large number of node pairs) exhibit the same increasing or decreasing trend over the course of the treatment. Specifically, the regions of strong negative correlation in the starting condition corresponded to regions of correlation increase, while regions of strong positive correlation in the starting condition corresponded to regions of correlation decrease (discounting on-diagonal terms, which of course exhibit no change by definition).

In the VISI treatment, there is less change overall (e.g. as quantified by the p-2 norm) although whether or not this difference is statistically significant cannot be easily demonstrated on one subject. Additionally, neighboring entries in the VISI correlation matrix shared similar values less often. In terms of what patterns still persist however, regions of strongly positive correlation in the starting condition seemed to favorably exhibit positive changes in correlation in cases where a change did occur, and strongly negative correlation regions seemed to favorably exhibit negative changes; both of these patterns were not as noticeable as in the CISI treatment. Rather than increase the sharpness of the small-scale features already present in the correlation matrix however, these changes predominantly "smoothed out" these regions, and therefore these changes largely occurred in the matrix cells that were the "exception to the norm" in that part of the matrix. A physical interpretation of this result will be provided with our graph theoretical representation of this correlation data, seen below.



(a) Change in correlation between the first and last 10 trials of P1 (CISI) condition



(b) Change in correlation between the first and last 10 trials of P2 (VISI) condition

Overall, in both the CISI and VISI treatments, there were dramatic and statistically significant changes in the correlation between the first and last ten trials. However, in the CISI treatment more change in the correlation occurs overall, and in a manner corresponding to shifts on larger feature scales. In particular, large regions of positively correlated values become much more negatively correlated, and large regions of negatively correlated values become much more positively correlated.

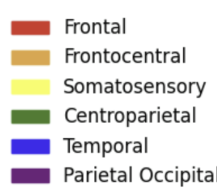
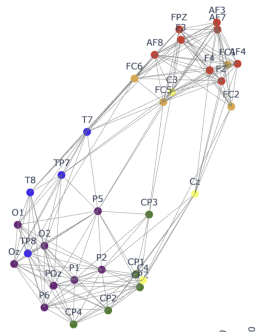
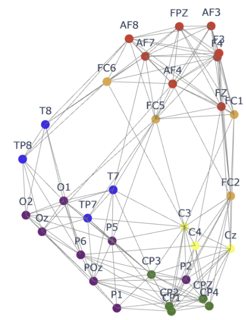


Figure 10: Color code for the graph nodes



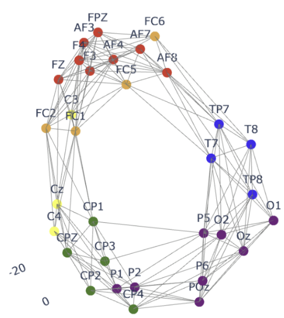
(a) First 10 CISI trials averaged



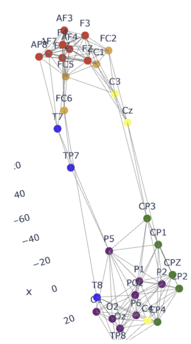
(b) Last 10 CISI trials averaged

Figure 11: Graph representation of P1 treatment (threshold = 0.4)

These figures show the graph network of the CISI condition. Disregarding any graph dynamics from the first to last 10 trials, we can immediately see that the locations of each electrode group generally match their anatomical locations. From the start to the end of the condition, we can see that there are not many significant changes, and the shape of the graph stays relatively similar. However, the frontal (position-wise) and rear groups come closer together, which implies that the weights between those regions have increased. A possible interpretation of this change is that there is some integration between these regions as the brain grew used to performing a constant task.



(a) First 10 VISI trials averaged



(b) Last 10 VISI trials averaged

Figure 12: Graph Representation of P2 (threshold = 0.3)

The P2 graphs display a much more pronounced change compared to those for P1 (Note: the differing thresholds are not significant). While the changes in the correlation matrix appeared to be more subtle, the reduction of small-scale features in the correlation matrix in the P2 case seem to produce a much greater change in the network structure than the strengthening of these small-scale features produced in the P1 case. In other words, increase of network sparsity has a greater impact on the network structure (and potentially its dynamics) compared to an increase of network complexity.

In the first 10 trials, we see a distinct ring shaped structure, which shows that correlations follow two general paths. By the end of the P2 condition, the ring has deformed into two unique paths, each defined by different brain regions. The first implication this has is that the brain has optimized information flow such that there are only a few paths that it is "willing" to go in, which are the most optimal (perhaps in speed, investigations into the lags along these paths may be in order). Observing both the CISI and VISI graphs, we also note that nodes in the somatosensory group are especially prone to changes in relative position, which could be explained by the brain adapting to the tactile stimulus in the experiment.

4 Discussion

As discussed with some of the earlier figures, we see that over a condition, there are significant changes that occur in the correlations, and these reflect themselves in the structure of the graph (most strongly seen in P2). Our results point towards a few possible findings. Firstly, we see that there is indeed some form of learning going on where the brain will adapt to most optimally send information given some repeated stimulus. Though perhaps trivial, our graphs also suggest that the brain is physically structured as to optimize connections, as we see that in the graphs, the relative positions of the electrodes in topological space moderately correspond to how they are actually positioned in the brain, which could suggest that the brain is already inherently structured to be optimized (although not a rigorous experimental finding, there were some instances where the graphs even preserved sidedness).

The unique approach we took enabled us to see patterns like loops or deformations in an extremely simple way, and allows us to visualize the complicated dynamics of the brain easily. In conjunction with observations of more "concrete" data, such as the correlation graphs, allows us to interpret the same behavior in different ways that each give different insights into the underlying processes. Some limitations that we faced were the size of the dataset (many of the metrics for "information" we aimed to use require extremely large number of samples), and trying to validate our results as processes like volume conduction diffuse the signals to various channels. In theory, a graph could be made by directly taking the raw correlation and assigning weights, and it would likely look relatively similar to our final results. However, properly filtering out noise and determining what data was important was the majority of our work.

One of the most immediate follow-up investigations that could be done is implementing directionality. The lags that we calculated already provide a notion of direction, and if we were to develop a cleaner method of finding the optimal lag, we could then investigate not just how information connects our network, but how it moves throughout the network. We suspect that although the current min-variance method favors signals that are most stable across time, for pairs with weak coupling, the variance curve across lags would be roughly flat and thus the argmin would pick noise. If we find optimal lags based on the max absolute mean across time or formulate a stronger metric like signal-to-noise ratio $\max|\operatorname{mean}(r)|/\operatorname{std}(r)$, we can look for lags where correlation is actually large and not fall to noisy pairs. Other intriguing investigations include the dynamics of the graph network in time more precisely than averaging over the first and last 10 trials as we did or finding steady states of our graph algorithm.

The acquisition of correlations could also be improved by using a larger dataset. For example, other subject data could be included to this end, or the EEG sample rate could be increased. With the increased signal-to-noise ratio gained through this change, we could perform STFT (short-time Fourier transform) such as modified discrete cosine transform to analyze the EEG data in the time-frequency domain. Correlation could then be filtered, or analyzed at different frequency bands.

5 Conclusion

Although there is some literature on this topic, graph-theoretic models of the brain still remain largely unexplored. Our work shows that topological approaches to model the brain can have interpretable and insightful results that cannot be fully grasped by other methods. With additional work, it is possible that this kind of approach could deepen our understanding of the inherent structure of the brain and how various components interact with each other.

References

[1] M Jacomy, T Venturini, S Heymann, and M Bastian. Forceatlas2, a continuous graph layout algorithm for handy network visualization designed for the gephi software. *PLOS One*, 2014.